BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 147 · Programming Languages and Translators

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How many input rows are needed for three independent Boolean inputs?

Q2. What does software copyright restrict?
A. Using any keyboard
B. Learning to code
C. Copying without permission
D. Saving your own work

Q3. 32 items are packed in boxes of 5. Use DIV and MOD to give the number of full boxes and leftover items.

Q4. Entry needs a valid ticket AND membership card. “AND is true if just one input is true.” Correct this. What happens with only a ticket?

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I Do · Read new knowledge

Key vocabulary

High-level language — Human-readable abstraction; generally portable with suitable runtime/translator support.

Low-level language — Close to processor instruction architecture, giving greater hardware control.

Compiler — Translates source code into executable machine or intermediate output before running the resulting program.

Interpreter — Processes/translates and executes code using an interpreter during execution.

New learning

High-level language: Human-readable abstraction; generally portable with suitable runtime/translator support.

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Worked example

A developer wants one program to run on different processors. Explain how language level and translators matter.

1. High-level code is less tied to a specific processor architecture.

2. A suitable translator/runtime must still exist for each target.

3. Low-level machine code is closely processor-specific.

4. Explain why a compiler/interpreter approach is chosen for the environment, not a guaranteed speed ranking.

We Do · Practise with guidance

Guided 1

Compare an interpreter and compiler in detecting errors.

Worked method: Do not claim all interpreters execute exactly one source line directly.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the size calculation for four two-byte characters.

```
characters = 4
bytesPerCharacter = 2
sizeBytes = characters + bytesPerCharacter
print(sizeBytes)
```

iterate

Question 1

Give one benefit of high-level languages. [2 marks]

Question 2

Compare compiler and interpreter at a broad level. [4 marks]

Question 3

Explain why machine code depends on processor architecture. [2 marks]

Exit Ticket · Answer before checking

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A high-level program still needs translation/runtime support to execute on a processor.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 8, because 2 to the power 3 is 8; list every combination exactly once.

Q2. C — Copying without permission

Q3. 32 DIV 5 = 6 full boxes; 32 MOD 5 = 2 left over.

Q4. AND is true only when both inputs are true. One valid ticket or card alone is not enough. OR needs at least one true input; NOT reverses a value.

I Do / We Do pages · Guided practice

Q1. Do not claim all interpreters execute exactly one source line directly.

You Do pages · Code First

```
characters = 4
bytesPerCharacter = 2
sizeBytes = characters * bytesPerCharacter
print(sizeBytes)
```

Expected result: 8

You Do pages · Independent questions

Q1. More readable and often easier to develop/maintain, or portable with suitable runtimes.

Q2. Compiler translates a program to executable/intermediate output; interpreter executes through an interpreting environment.

Q3. Instructions use a machine-specific instruction set understood by that CPU.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: